

Question 5

Q: A surveillance system requires real-time image processing. Classify the level of Computer Vision involved and justify your choice.

Theoretical Concept: Levels of Computer Vision

A surveillance system involves multiple levels:

1. **Low-Level CV:** Image acquisition, noise reduction, and contrast enhancement.
2. **Mid-Level CV:** Motion detection (background subtraction) and edge detection.
3. **High-Level CV:** Object recognition (identifying a person or vehicle) and semantic understanding (detecting suspicious behavior).

OpenCV Python Solution

```
# =====  
# AIM:  
# Demonstrate High-Level Computer Vision using Human Detection  
# in a Surveillance Image (Google Colab)  
# =====  
  
import cv2  
import matplotlib.pyplot as plt  
from google.colab import files  
  
# Upload surveillance image  
print("Upload a surveillance image")  
uploaded = files.upload()  
  
image_name = next(iter(uploaded))  
  
# Read image  
img = cv2.imread(image_name)  
  
# Resize image for better processing
```

```

img = cv2.resize(img, (700, 500))

# Initialize HOG Person Detector
hog = cv2.HOGDescriptor()
hog.setSVMDetector(cv2.HOGDescriptor_getDefaultPeopleDetector())

# Detect people
boxes, weights = hog.detectMultiScale(
    img,
    winStride=(8,8),
    padding=(16,16),
    scale=1.05
)

count = 0

# Draw bounding boxes
for (x, y, w, h) in boxes:
    count += 1
    cv2.rectangle(img, (x, y), (x+w, y+h), (255,0,0), 2)
    cv2.putText(img,
        f"Person {count}",
        (x, y-8),
        cv2.FONT_HERSHEY_SIMPLEX,
        0.6,
        (255,0,0),
        2)

# Display image
plt.figure(figsize=(10,7))
plt.imshow(cv2.cvtColor(img, cv2.COLOR_BGR2RGB))
plt.title("Surveillance Human Detection")
plt.axis("off")
plt.show()

# Result
print("="*45)
print("SURVEILLANCE REPORT")
print("="*45)
print("Humans Detected :", count)

```

```
if count > 0:
    print("Computer Vision Level : HIGH-LEVEL")
    print("Reason : The system automatically detects")
    print("          and recognizes people in the scene.")
else:
    print("No humans detected.")
```

Expected Output

The output will display a completely black image with a stark white square in the center. The algorithm successfully ignored the static background and mathematically isolated the 'moving' object.